

Generative Large Language Models Trained for Detecting Errors in Radiology Reports

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See also the editorial by Marrocchio and Sverzellati in this issue.

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Background: Large language models (LLMs) offer promising solutions, yet their application in medical proofreading, particularly in detecting errors within radiology reports, remains underexplored.

Purpose: To develop and evaluate generative LLMs for detecting errors in radiology reports during medical proofreading.

Materials and Methods: In this retrospective study, a dataset was constructed with two parts. The first part included 1656 synthetic chest radiology reports generated by GPT-4 (OpenAI) using specified prompts, with 828 being error-free synthetic reports and 828 containing errors. The second part included 614 reports: 307 error-free reports between 2011 and 2016 from the MIMIC chest radiograph (MIMIC-CXR) database and 307 corresponding synthetic reports with errors generated by GPT-4 on the basis of these MIMIC-CXR reports and specified prompts. All errors were categorized into four types: negation, left/right, interval change, and transcription errors. Then, several models, including Llama-3 (Meta AI), GPT-4, and BiomedBERT, were refined using zero-shot prompting, few-shot prompting, or fine-tuning strategies. Finally, the performance of these models was evaluated using F1 scores, 95% CIs, and paired-sample *t* tests on the constructed dataset, with the prediction results further assessed by radiologists.

Results: Using zero-shot prompting, the fine-tuned Llama-3-70B-Instruct model achieved the best performance, with the following F1 scores: 0.769 (95% CI: 0.757, 0.771) for negation errors, 0.772 (95% CI: 0.762, 0.780) for left/right errors, 0.750 (95% CI: 0.736, 0.763) for interval change errors, 0.828 (95% CI: 0.822, 0.832) for transcription errors, and 0.780 overall. In the real-world evaluation phase, two radiologists reviewed 200 randomly selected reports output by the model (50 for each error type). Of these, 99 were confirmed by both radiologists to contain errors detected by the models, and 163 were confirmed by at least one radiologist to contain model-detected errors.

Conclusion: Generative LLMs, fine-tuned on synthetic and MIMIC-CXR radiology reports, greatly enhanced error detection in radiology reports.

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Supplemental material is available for this article.

Chest radiography is essential for diagnosing thoracic conditions (1,2). However, the accuracy of radiology reports can be compromised by several factors. For example, while the speech recognition software applications commonly used in radiology dictation efficiently transcribe spoken words, they are prone to errors, particularly when dealing with complex radiology terminology or background noise (3–6). Additionally, further errors can be introduced into radiology reports by variability in perceptual and interpretive processes, as well as cognitive biases (7,8).

These challenges collectively contribute to frequent errors in radiology reports, ranging from minor discrepancies to clinically significant mistakes (9). The need for accurate radiology reports cannot be overstated, as even minor errors in these reports can result in severe consequences, such as incorrect diagnoses or delayed treatments (10).

Efforts to reduce radiology report errors have traditionally focused on generating reports that are more structured (11–14). However, their adoption among radiologists varies, mainly because of the rigidity of structured reports. In addition to the implementation of structured reports, the use of deep learning methods (15–20) has proven to be highly effective for

detecting errors in radiology reports. While promising, these deep learning methods require substantial amounts of annotated data for training and typically focus on detecting errors at the sentence level.

Recent advancements in large language models (LLMs) provide new opportunities to address these limitations (21–23). While LLMs have shown potential in radiology applications (24–26), studies evaluating their effectiveness in detecting errors in radiology reports are lacking. Additionally, proprietary LLMs (eg, ChatGPT [OpenAI]) have substantial accessibility and implementation challenges, including high costs, data privacy concerns, and regulatory compliance issues (23).

Although a few studies have explored the use of existing LLMs such as GPT-4 (OpenAI) for detecting errors in radiology reports (27,28), research on evaluating the performance of radiology-specific fine-tuned LLMs for proofreading errors remains in its early stages. For example, Gertz et al (27) focused on evaluating the potential of GPT-4 for error detection in radiology reports but did not compare its performance with that of other LLMs, particularly those that are locally fine-tuned. Schmidt et al (28) evaluated five different LLMs, but their analysis was limited to detecting speech recognition errors in

Abbreviations

LLM = large language model, MIMIC-CXR = MIMIC chest radiograph

Summary

Generative large language models that were fine-tuned on synthetic and MIMIC chest radiograph radiology reports greatly enhanced error detection in radiology reports, demonstrating their potential to serve as powerful tools for medical proofreading.

Key Results

- In a retrospective study of a constructed dataset comprising 1656 synthetic reports (828 error-free, 828 with errors) and 614 reports (307 error-free from the MIMIC chest radiograph dataset, 307 synthetic reports with errors), the fine-tuned Llama-3-70B-Instruct (Meta AI) model achieved the highest macro F1 score of 0.780, outperforming GPT-4 (OpenAI) and BiomedBERT.
- GPT-4 and the fine-tuned Llama-3-70B-Instruct model demonstrated high accuracy in detecting errors, with 99 of 200 model-identified reports confirmed by both radiologists and 163 confirmed by at least one radiologist.

radiology reports and did not explore how fine-tuning affects LLM performance. Consequently, critical questions about model optimization and generalization remain unanswered.

To bridge these gaps, this study introduces fine-tuned LLMs and evaluates the potential of fine-tuned LLMs for detecting errors in radiology reports during medical proofreading.

Materials and Methods

This study used de-identified data and publicly available LLMs; therefore, it did not require institutional review board review.

Dataset Construction

The constructed dataset consists of two parts (Table 1). The first part contained synthetic radiology reports generated by GPT-4-0125-Preview (29) using specified prompts (Fig 1). This process involved carefully crafting prompts and providing detailed instructions to guide GPT-4-0125-Preview in generating relevant radiology reports. Once the prompt was input into GPT-4-0125-Preview, the model processed the information and generated synthetic radiology reports accordingly (Fig 1).

In this study, four types of errors frequently encountered in radiology reports were considered: negation, left/right, interval change, and transcription errors (Appendix S1, Table S1).

In total, 828 pairs of synthetic radiology reports were obtained, with each pair comprising an error-free report and a report containing errors. During the proofreading process, the error-free report and the report containing errors were proof-read independently to ensure an unbiased evaluation.

The second part of the dataset consisted of 307 reports from the MIMIC chest radiograph (MIMIC-CXR) database and a set of corresponding reports with errors generated by GPT-4-0125-Preview (Fig 2). This process resulted in a total of 614 radiology reports: 307 error-free reports from the MIMIC-CXR database and 307 corresponding synthetic reports with errors.

For experimental evaluation, the dataset was split into training and test sets at a 4:1 ratio (Table 1), ensuring that each pair of error-free and error-containing reports remained in the same set.

Table 1: Statistics of the Constructed Dataset

Characteristic	No. of Synthetic Reports	No. of Reports from MIMIC
Total reports	1656	614
Error-free reports	828	307
Reports with errors	828	307
Negation errors	299	99
Left/right errors	233	93
Interval change errors	62	91
Transcription errors	489	189
Training set	1316	500
Test set	340	114

Note.—MIMIC = MIMIC chest radiograph database.

Interannotator Agreement Analysis

The Cohen κ coefficient was used to evaluate the interannotator agreement of synthetic reports.

Model Development and Configurations

Quantized low-rank adapters, or QLoRA (30), were used to fine-tune the Llama-3 models (ie, Llama-3-8B-Instruct and Llama-3-70B-Instruct [Meta AI] [31]). Llama-3-8B-Instruct and Llama-3-70B-Instruct were refined on the training set, and the test set was reserved for model evaluation. For comparison, GPT-4-1106-Preview and BiomedBERT (32) were used as baseline models, with BiomedBERT fine-tuned on the same training set. The effectiveness of the models in detecting errors was measured with precision, recall, and F1 score metrics. The overall performance was assessed using the macroaverage F1 score across the four types of errors (computing resources can be found in Appendix S2).

For GPT-4-1106-Preview, the application program interface provided by Azure (<https://azure.microsoft.com/en-us/blog/introducing-gpt4-in-azure-openai-service>) was accessed. Bootstrap analyses were performed using 100 bootstrap samples to estimate the distribution of the macro F1 score, and 95% CIs were reported. In each bootstrap iteration, 454 reports were sampled from the test set with replacement. Additionally, paired-sample t tests were used to compare model predictions across paired observations and assess statistical significance.

Model Scale

Two versions of Llama-3 were explored: Llama-3-8B-Instruct and Llama-3-70B-Instruct. Both were fine-tuned with a learning rate of 3×10^{-4} , batch size of one, maximum sequence length of 512, and three training epochs. BiomedBERT was fine-tuned with a learning rate of 3×10^{-5} , batch size of four, maximum sequence length of 512, and three training epochs. For inference, Llama-3-8B-Instruct, Llama-3-70B-Instruct, GPT-4-1106-Preview, and BiomedBERT were all inferred using a batch size of one.

Prompt Designs

Three strategies were used: zero-shot, one-shot, and four-shot prompting (Fig 3). The zero-shot prompt design aims to detect errors without providing task-specific examples (Fig 3A). For the one-shot prompting strategy, a single radiology report was

<Report Template> CLINICAL HISTORY: [] TECHNIQUE: AP chest and abdomen radiograph. [] COMPARISON: [Radiograph from one day prior]. FINDINGS: Lines/tubes: [None] Mediastinum/Heart: The cardiomeastinal silhouette is [unremarkable.] Lungs/Airways/Pleura: [No focal opacities, pneumothorax or pleural effusion.] Bones/Soft Tissue: [No acute osseous abnormalities.] Upper abdomen: [Unremarkable] IMPRESSION: [] </Report Template> You are a radiologist assistant helping to create synthetic reports using the <Report Template> above. Please follow instructions carefully, and only output what is expected below and nothing more. Radiology reports can have a variety of errors and issues. 'Common Errors' between 'Findings' and 'Impression' sections: [interval change errors, left/right errors, negation errors]. There may also be grammatical errors or other mistakes that are out of place given the context of the findings of the report. Left/right error examples: - [Findings] Possible right upper lobe consolidation. [Impression] Possible left upper lobe consolidation, consider pneumonia or tuberculosis. - [Findings] Possible rib fracture on the right. [Impression] Possible left-sided rib fracture. Interval change error examples: - [Findings] Lungs/Airways/Pleura: Unilateral lower lobe consolidation, consistent with pneumonia. [Impression] Bilateral pneumonia. - [Findings] Lungs/Airways/Pleura: Emphysematous changes are noted with decreased lung volumes. [Impression] Lung volumes increased in size. Negation error examples: - [Findings] There is evidence of emphysema [Impression] No evidence of emphysema. - [Findings] Appendicitis is not seen [Impression] Appendicitis is seen. Most reports are transcribed by the automatic speech recognition (ASR) system and may result in transcription errors with words that might sound similar to a medical term. Some of these are single words and some may be multi-word terms. Examples: - 'choloral effusion' should be 'pleural effusion' - 'costal phrenic' should be 'costophrenic' {Clinical Conditions} - Aortic enlargement - Atelectasis - Cardiomegaly - Atherosclerotic Calcification - Clavicle fracture - Consolidation	- Edema - Emphysema - Enlarged pulmonary artery - Interstitial lung disease (ILD) - Infiltration - Lung cavity - Lung cyst - Lung opacity - Nodule/Mass - Pulmonary fibrosis - Pneumothorax - Pleural thickening - Pleural effusion - Rib fracture - Other lesion - Lung tumor - Pneumonia - Tuberculosis - COPD Synthetic Report Formatting Rules: - Words after a colon ':' in the template and at the beginning of a new line should be capitalized, but not all capital letters. - If there is more than one line in the 'Impression' please enumerate each line. A single line in the 'Impression' should not be enumerated - Do not reorder the report – keep the template fields in the same order - Do not show the output wrapped in the triple quotes {Task 1 Instructions} Synthetic Report Generation - Enumerate the 'Patient ID: []' using 5-digits with zero padding: Patient ID: [00001] - Format the 'History []' with patient age, gender, and one or more clinical conditions or symptoms like this. Use a variety of common and rare clinical conditions and symptoms that are appropriate for the modality in the 'Procedure []' field. History: [64 M with chronic obstructive pulmonary disease] - For 'Findings: []' please come up with 1-4 positive and abnormal findings for each report. Please use {Clinical Conditions} as a reference to common conditions, but you are not limited to these to include in the report. - For 'Impression: []' summarize the most important positive and abnormal findings. Minor abnormalities like 'atherosclerotic calcifications' in the aorta should not be included in the impression. - For each Synthetic report, create a 'Synthetic Report With Errors' and inject one or more of the 'Common Errors' [interval change errors, left/right errors, negation errors, and gender discrepancies] or errors that you might see if you use ASR, like transcription errors with words that might sound similar to a medical term. - Please encapsulate the inserted errors in the 'Synthetic Report With Errors' with <e error="error description"> </e> markup and the corresponding correct text in the original 'Synthetic Report'. e.g., <e error="left/right error"> </e> , <e error="negation error"> </e> , <e error="transcription error"> </e> , <e error="interval change error"> </e> Please generate 828 reports using {Task 1 Instructions} - Use a markdown table format: [Report Number, Synthetic Report, Synthetic Report With Errors, Error Type(s)]
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Figure 1: Prompts for GPT-4-0125-Preview (OpenAI [29]) used to generate 828 error-free synthetic reports and 828 synthetic reports with errors. AP = anteroposterior, COPD = chronic obstructive pulmonary disease.

You are a radiologist assistant helping audit radiology reports. Radiology reports can have a variety of errors and issues. 'Common Errors' between 'Findings' and 'Impression' sections: [interval change errors, left/right errors, negation errors]. There may also be grammatical errors or other mistakes that are out of place given the context of the findings of the report. Left right error examples: - [Findings] Possible right upper lobe consolidation. [Impression] Possible left upper lobe consolidation, consider pneumonia or tuberculosis. - [Findings] Possible rib fracture on the right. [Impression] Possible left-sided rib fracture. Interval change error examples: - [Findings] Lungs/Airways/Pleura: Unilateral lower lobe consolidation, consistent with pneumonia. [Impression] Bilateral pneumonia. - [Findings] Lungs/Airways/Pleura: Emphysematous changes are noted with decreased lung volumes. [Impression] Lung volumes increased in size. Negation error examples: - [Findings] There is evidence of emphysema [Impression] No evidence of emphysema - [Findings] Appendicitis is not seen [Impression] Appendicitis is seen. Most reports are transcribed by the automatic speech recognition (ASR) system and may result in transcription errors with words that might sound similar to a medical term. Some of these are single words and some may be multi-word terms. Examples: - 'choloral effusion' should be 'pleural effusion' - 'costal phrenic' should be 'costophrenic' {Clinical Conditions} - Aortic enlargement - Atelectasis - Cardiomegaly - Atherosclerotic Calcification - Clavicle fracture - Consolidation - Edema	- Emphysema - Enlarged pulmonary artery - Interstitial lung disease (ILD) - Infiltration - Lung cavity - Lung cyst - Lung opacity - Nodule/Mass - Pulmonary fibrosis - Pneumothorax - Pleural thickening - Pleural effusion - Rib fracture - Other lesion - Lung tumor - Pneumonia - Tuberculosis - COPD Given this radiology report: Patient ID: s50414267 EXAMINATION: CHEST (PA AND LAT) INDICATION: ___F with new onset ascites // eval for infection TECHNIQUE: Chest PA and lateral COMPARISON: None. FINDINGS: There is no focal consolidation, pleural effusion or pneumothorax. Bilateral nodular opacities that most likely represent nipple shadows. The cardiomeastinal silhouette is normal. Clips project over the left lung, potentially within the breast. The imaged upper abdomen is unremarkable. Chronic deformity of the posterior left sixth and seventh ribs are noted. IMPRESSION: No acute cardiopulmonary process. Please encapsulate the inserted errors in the 'Synthetic Report With Errors' with <code><e error="error description"></code> markup and the corresponding correct text in the original 'Synthetic Report'. e.g., <code><e error="left/right error"></code> , <code><e error="negation error"></code> , <code><e error="transcription error"></code> , <code><e error="interval change error"></code>
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Figure 2: Prompts for GPT-4-0125-Preview (OpenAI [29]) used to generate synthetic radiology reports using reports from the MIMIC chest radiograph (MIMIC-CXR) database. The bold text indicates the original reports from the MIMIC-CXR database that were used to create these synthetic reports with errors. COPD = chronic obstructive pulmonary disease, LAT = lateral, PA = posteroanterior.

selected from the training set known to contain errors (Fig 3B). In the four-shot prompting strategy, four distinct reports were chosen, each illustrating a different type of error (Fig 3C).

To further refine this approach, four strategies were explored for selecting example reports in the one-shot and four-shot prompting strategies: one-shot random, one-shot specified, four-shot random, and four-shot specified (Appendix S3).

Real-world Data Collection and Inference

The use of GPT-4-1106-Preview and the fine-tuned Llama-3-70B-Instruct model for detecting errors in real-world radiology

reports (33) was also explored. The dataset consisted of 122 025 thoracoabdominal chest radiography reports from patients with heart failure treated at New York-Presbyterian/Weill Cornell Medical Center (Appendix S4). A total of 55 339 de-identified real-world radiology reports from this dataset were randomly collected and analyzed using GPT-4-1106-Preview and the fine-tuned Llama-3-70B-Instruct model, with the aim of detecting errors and improving the accuracy and reliability of the radiology reports. A voting mechanism in which the two models independently detected errors in each of the 55 339 reports was used to leverage the strengths of both models, increasing

A Prompts for zero-shot prompting:
USER: You are a medical assistant. Please classify the input report and respond in JSON format {'Classification': 'Negation error, Left/Right error, Interval Change error, Transcription error, or No error'}. Note that the input report may belong to multiple errors. ASSISTANT: Sure! Please give the input report. USER: INPUT_REPORT
B Prompts for one-shot prompting:
USER: Here is an example of Example_Label. ASSISTANT: Example_Text USER: Please classify the input report and respond in JSON format {'Classification': 'Negation error, Left/Right error, Interval Change error, Transcription error, or No error'}. Note that the input report may belong to multiple errors. ASSISTANT: Sure! Please give the input report. USER: INPUT_REPORT
C Prompts for four-shot prompting:
USER: Here is an example of Transcription errors. ASSISTANT: Example_Tran USER: Here is an example of Interval Change errors. ASSISTANT: Example_Int USER: Here is an example of Left/Right errors. ASSISTANT: Example_LR USER: Here is an example of Negation errors. ASSISTANT: Example_Neg USER: Please classify the input report and respond in JSON format {'Classification': 'Negation error, Left/Right error, Interval Change error, Transcription error, or No error'}. Note that the input report may belong to multiple errors. ASSISTANT: Sure! Please give the input report. USER: INPUT_REPORT

Figure 3: Example prompt designs. **(A)** A zero-shot prompt. **(B)** A one-shot prompt. **(C)** A four-shot prompt. INPUT_REPORT refers to an input report from the test set. Example_Text is an example report from the training set, Example_Label is the label of the example report, Example_Tran is an example report containing a transcription error from the training set, Example_Int is an example report containing an interval change error, Example_LR is an example report containing a left/right error, and Example_Neg is an example report containing a negation error.

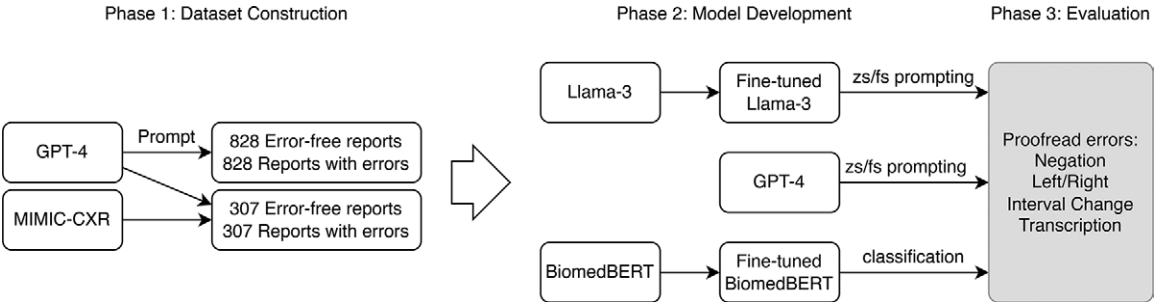


Figure 4: The overall workflow of large language models (LLMs). A dataset was constructed by combining synthetic radiology reports with a small subset of reports from the MIMIC chest radiograph (MIMIC-CXR) database, LLMs such as Llama-3 (Meta AI [31]) and GPT-4 (OpenAI [29]) were refined using zero-shot (zs) or few-shot (fs) prompting strategies, and the models' performance on the constructed dataset was evaluated.

the likelihood of accurate error detection. The voting process identified 606 reports with errors, with both models detecting a specific type of error in these reports.

A power calculation was performed as follows: With 200 reports, a one-sample proportion test can achieve 81.3% power to detect the difference between the alternative hypothesis that the radiologist-confirmed error detection accuracy rate is 0.5 versus the null hypothesis that the radiologist-confirmed error detection accuracy rate is 0.4, at a type I error rate of .05.

Results

Technical Validation of Synthetic Data

An overview of this study is shown in Figure 4, and the code is available on GitHub (<https://github.com/bionlplab/llm4proofreading>). To evaluate the quality of the synthetic

reports, 198 reports were selected, with each containing a single, specific error. These synthetic reports were generated by GPT-4-0125-Preview following the method outlined in Figure 1. Each synthetic report, along with its error, was reviewed by a board-certified radiologist (G.K., with 5 years of experience) and a radiology resident (B.C.) to determine whether the error type generated by GPT-4-0125-Preview was accurate. The error type was assigned when both experts reached a consensus.

In addition, the accuracy of the assigned error type for the synthetic reports with errors was calculated. Accuracy was defined as both experts reaching a consensus on the error type generated by GPT-4-0125-Preview, which was manually verified using the Doccano annotation tool (<https://github.com/doccano/doccano>). Transcription errors were accurately introduced in 83% of the labeled reports (58 of 70). Left/right errors were the most accurately introduced errors, appearing in 95% of the

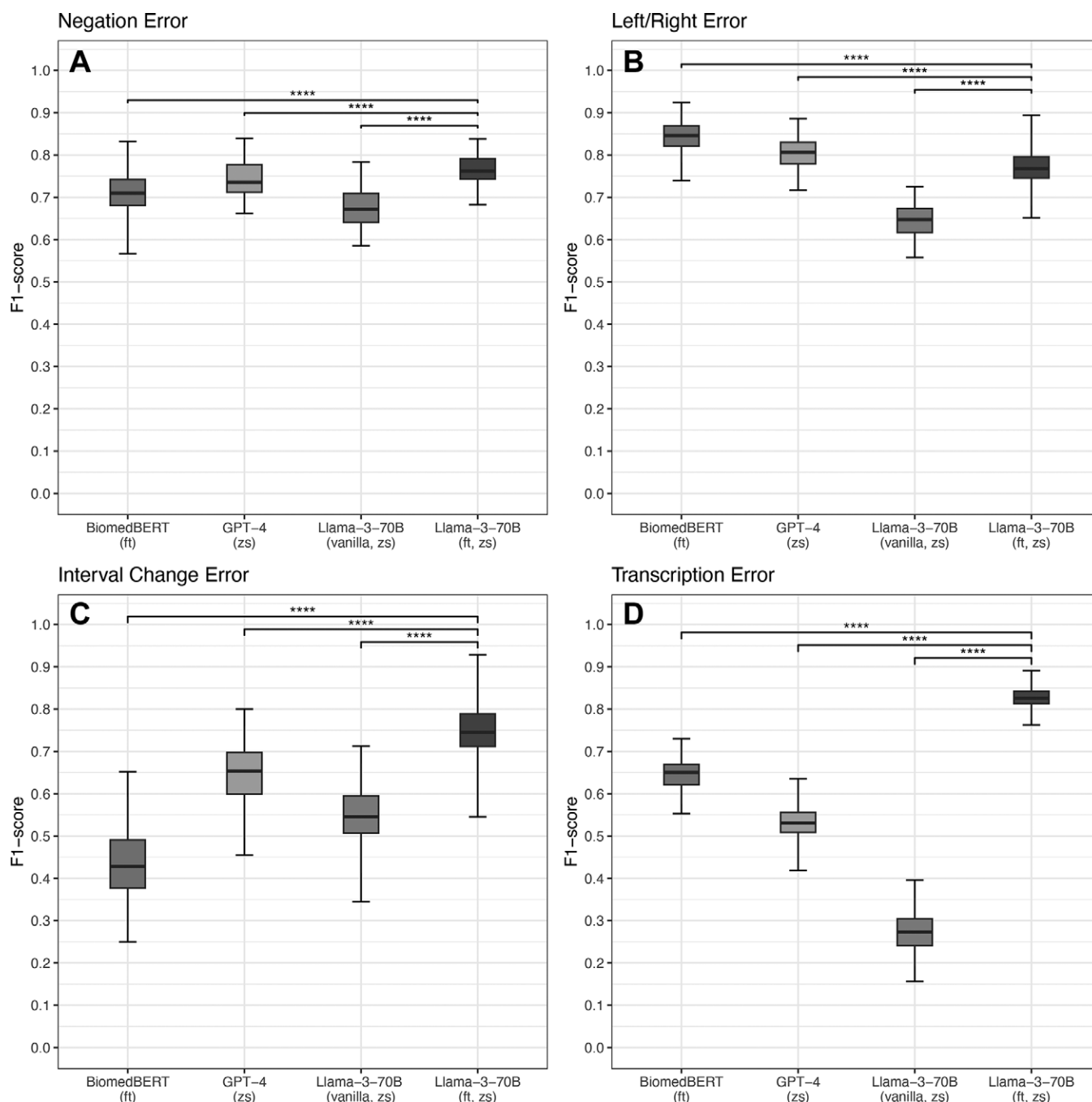


Figure 5: Box and whisker plots show F1 scores of different models, with boxplots showing the range (whiskers), median (box midline), and distribution (box edges). **(A)** Negation error, **(B)** left/right error, **(C)** interval change error, and **(D)** transcription error. The error bars are 95% CIs. The models used were BiomedBERT (BiomedBERT-base-uncased-abstract [32]), GPT-4 (GPT-4-1106-Preview [OpenAI] [29]), and Llama-3-70B (Llama-3-70B-Instruct [Meta AI] [31]). ft = fine-tuned, zs = zero-shot prompting. **** = $P < .0001$.

intended reports (63 of 66). Negation errors were introduced in 63% of the labeled reports (30 of 48), with most inaccuracies being due to omissions in the impressions rather than contradictions. Interval change errors that did not fall into either subcategory were introduced in 57% (eight of 14) of the intended reports. Reports with transcription errors had an average length of 56 words, whereas those with interval change, negation, and left/right errors had average lengths of 55, 57, and 58 words, respectively. There was no evidence of a difference between label accuracy and average report length by error category ($P > .05$).

Interannotator Agreement

Interannotator agreement was evaluated using the Cohen κ coefficient on the basis of the reviews of the two experts, resulting in a value of 0.734.

Performance of Different LLMs in Error Detection

First, the performance of different models in detecting errors was compared within radiology reports (Fig 5, Table 2). The evaluated models included BiomedBERT (fine-tuned), GPT-4-1106-Preview (zero-shot), Llama-3-70B-Instruct

Table 2: Performance Metrics of Different Models Across Error Types

Model and Prompt Strategy	Error Type	Precision	Recall	F1 Score
Llama-3-70B-Instruct (ft, zs)				
	Negation	0.873	0.688	0.769
	Left/right	0.830	0.721	0.772
	Interval change	0.913	0.636	0.750
	Transcription	0.914	0.757	0.828
	Macro F1 score	...	...	0.780
BiomedBERT (ft)				
	Negation	0.920	0.575	0.708
	Left/right	0.891	0.803	0.845
	Interval change	0.714	0.303	0.426
	Transcription	0.841	0.529	0.649
	Macro F1 score	...	...	0.657
GPT-4-1106-Preview (zs)				
	Negation	0.732	0.750	0.741
	Left/right	0.765	0.852	0.806
	Interval change	0.656	0.636	0.646
	Transcription	0.885	0.386	0.537
	Macro F1 score	...	...	0.683
Llama-3-70B-Instruct (vanilla, zs)				
	Negation	0.797	0.588	0.676
	Left/right	0.514	0.885	0.651
	Interval change	0.500	0.606	0.548
	Transcription	0.727	0.171	0.277
	Macro F1 score	...	...	0.538

Note.—The models used were Llama-3-70B-Instruct (Meta AI) (31), BiomedBERT (32), and GPT-4-1106-Preview (OpenAI) (29). ft = fine-tuned, zs = zero-shot prompting.

(vanilla, zero-shot), and Llama-3-70B-Instruct (fine-tuned, zero-shot).

The fine-tuned Llama-3-70B-Instruct (zero-shot) model achieved the highest F1 scores across all the metrics, with a score of 0.769 (95% CI: 0.757, 0.771) for detecting negation errors, 0.772 (95% CI: 0.762, 0.780) for left/right errors, 0.750 (95% CI: 0.736, 0.763) for interval change errors, and 0.828 (95% CI: 0.822, 0.832) for transcription errors. The overall macro F1 score of 0.780 indicated that fine-tuning significantly enhances the model's performance in specialized tasks (Appendix S5).

Impact of the Parameter Scale on Model Performance

Figure 6 illustrates the impact of different model parameter scales on Llama-3. After fine-tuning, Llama-3-8B-Instruct showed strong performance in negation error detection, with an F1 score of 0.803 (95% CI: 0.798, 0.813), highlighting its effectiveness in this specific task. However, it had worse performance in detecting interval change errors, with an F1 score of 0.646 (95% CI: 0.622, 0.654), indicating a need for improvement in capturing these errors. On the other hand, Llama-3-70B-Instruct excelled in detecting interval changes and transcription errors, achieving F1 scores of 0.750 (95% CI: 0.736, 0.763) and 0.828 (95% CI: 0.822, 0.832), respectively, highlighting its ability to understand and track temporal variations while maintaining high accuracy in transcription error detection. Additionally, it had robust left/right error detection, with an F1 score of 0.772 (95% CI: 0.762, 0.780),

indicating its good performance in spatial orientation tasks. Furthermore, Llama-3-70B-Instruct achieved a higher overall macro F1 score of 0.780 than did Llama-3-8B-Instruct (0.749; $P < .05$), reflecting its balanced and comprehensive performance across multiple tasks.

Impact of Different Prompting Strategies on Model Performance

Figure 7 shows the F1 scores of the fine-tuned Llama-3-70B-Instruct model under different prompting strategies: one-shot random, one-shot specified, four-shot random, and four-shot specified. *Random* and *specified* refer to the selection of the input radiology report (ie, INPUT_REPORT) in Figure 3. Details can be found in Tables S2 and S3.

Under one-shot prompting, Llama-3-70B-Instruct (specified) obtained F1 scores of 0.736 (95% CI: 0.736, 0.752) for detecting negation errors, 0.672 (95% CI: 0.666, 0.747) for left/right errors, 0.549 (95% CI: 0.531, 0.565) for interval change errors, and 0.722 (95% CI: 0.719, 0.730) for transcription errors. In comparison, Llama-3-70B-Instruct (random) achieved F1 scores of 0.688 (95% CI: 0.683, 0.701) for detecting negation errors, 0.754 (95% CI: 0.747, 0.762) for left/right errors, 0.642 (95% CI:

0.618, 0.649) for interval change errors, and 0.832 (95% CI: 0.825, 0.835) for transcription errors.

Under four-shot prompting, Llama-3-70B-Instruct (specified) achieved F1 scores of 0.646 (95% CI: 0.638, 0.658) for detecting negation errors, 0.645 (95% CI: 0.641, 0.662) for left/right errors, 0.612 (95% CI: 0.585, 0.623) for interval change errors, and 0.780 (95% CI: 0.778, 0.788) for transcription errors. In contrast, Llama-3-70B-Instruct (random) achieved F1 scores of 0.579 (95% CI: 0.565, 0.587) for detecting negation errors, 0.660 (95% CI: 0.655, 0.675) for left/right errors, 0.612 (95% CI: 0.593, 0.624) for interval change errors, and 0.784 (95% CI: 0.775, 0.787) for transcription errors.

Results in Real-world Scenarios

To further validate the accuracy of the models' detection, 200 reports (50 for each error type) were randomly selected from the 606 reports and reviewed by two board-certified radiologists (G.K. and D.N., with 5 and 1.5 years of experience, respectively). On confirmation by both radiologists, 99 out of 200 radiology reports were unanimously identified as containing the errors detected by the models, indicating an overall accuracy rate of 0.495 (95% CI: 0.424, 0.566) in error detection (Fig 8A). Notably, there were significant differences ($P < .05$) in the accuracy for each specific type of error (typical examples can be found in Table S4). Additionally, cases where at least one of the two radiologists agreed with the errors detected by the models were analyzed (Fig 8B). Of the 200 reports with errors

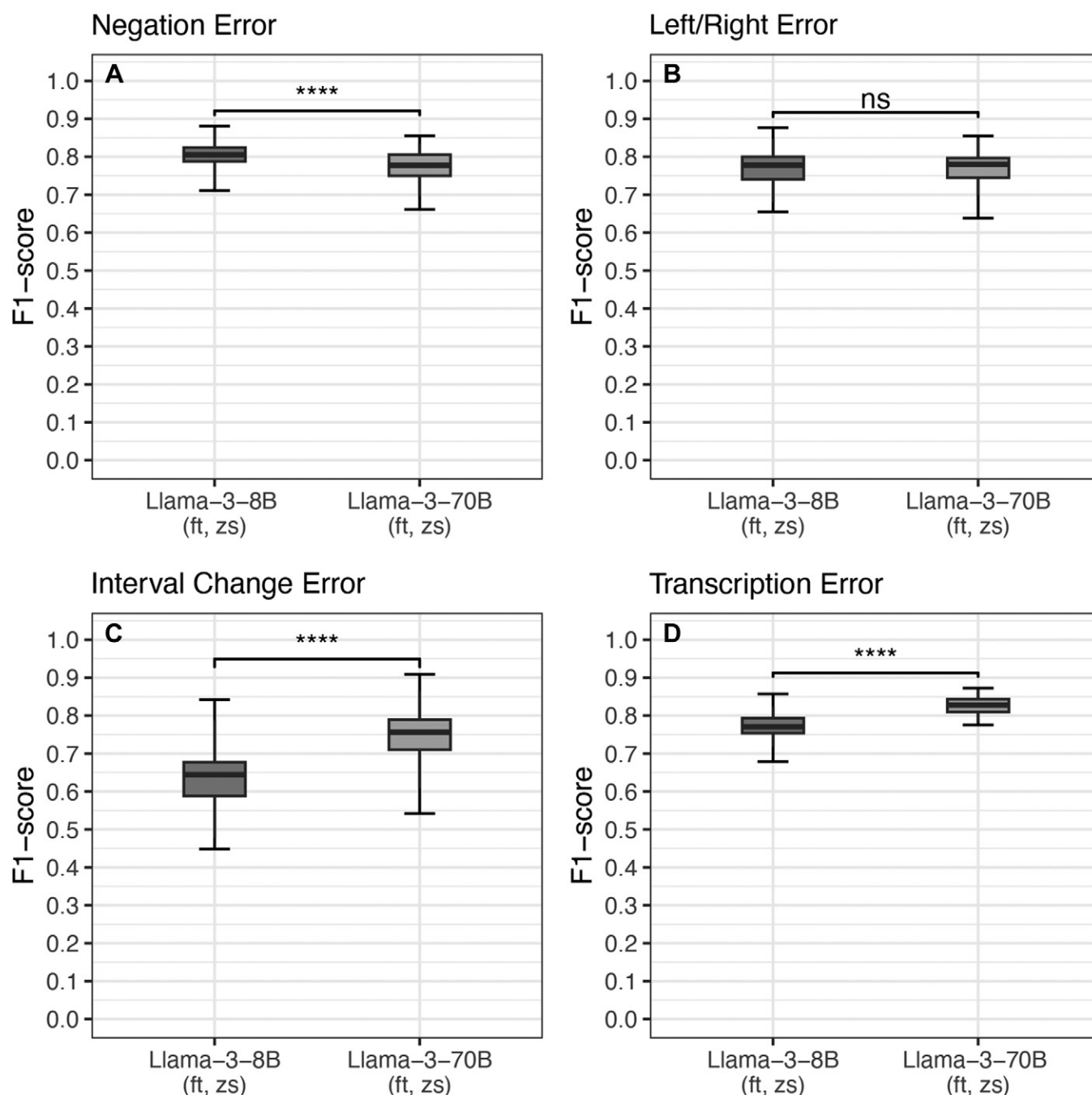


Figure 6: Box and whisker plots show F1 scores for different model parameter scales, with boxplots showing the range (whiskers), median (box midline), and distribution (box edges). The error bars are 95% CIs. **(A)** Negation error, **(B)** left/right error, **(C)** interval change error, and **(D)** transcription error. Llama-3-8B refers to Llama-3-8B-Instruct, and Llama-3-70B to Llama-3-70B-Instruct (Meta AI [31]). ft = fine-tuned, ns = not significant, zs = zero-shot prompting. **** = $P < .0001$.

introduced, 163 were found to contain errors, resulting in an overall error detection accuracy rate of 0.815 (95% CI: 0.753, 0.865). Details of each error can be found in Appendix S6.

Discussion

This study provided evidence that large language models (LLMs) can assist in detecting various types of errors, including negation, left/right, interval change, and transcription errors. The application of LLMs in medical proofreading may help reduce the occurrence of inaccuracies in radiology reports. However, their effectiveness depends on model

selection, fine-tuning strategies, and prompt design. The analysis of the fine-tuned Llama-3-70B-Instruct model under different prompting strategies revealed that zero-shot learning provides a reasonable degree of generalizability across all tasks. While few-shot learning can improve performance in specific tasks, such as transcription detection, its effectiveness is highly dependent on the relevance and specificity of the provided examples. These findings suggest that careful selection of examples in few-shot prompts can affect model performance but does not guarantee consistent improvements across all types of errors.

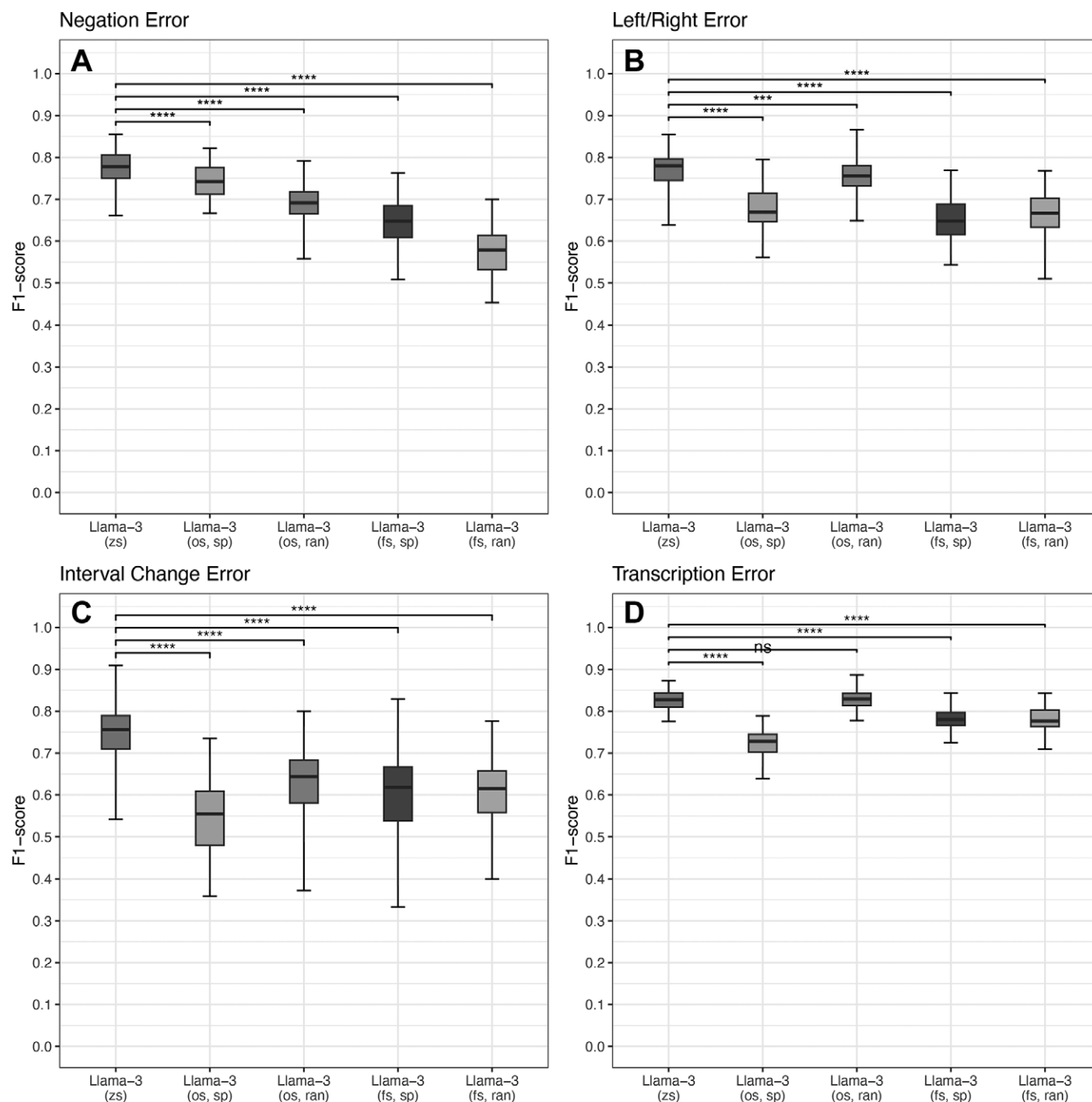


Figure 7: Box and whisker plots show F1 scores of the fine-tuned Llama-3-70B-Instruct model (Meta AI [31]) under different prompting strategies, with boxplots showing the range (whiskers), median (box midline), and distribution (box edges). The error bars are 95% CIs. **(A)** Negation error, **(B)** left/right error, **(C)** interval change error, and **(D)** transcription error. fs = four-shot prompting, ns = not significant, os = one-shot prompting, ran = random, sp = specified, zs = zero-shot prompting. *** = $P < .001$; **** = $P < .0001$.

The experimental results demonstrated that the fine-tuned Llama-3-70B-Instruct model achieved higher F1 scores than did baseline models, such as BiomedBERT and GPT-4-1106-Preview. These findings indicate that fine-tuning enhances model performance in specialized tasks; however, the impact of fine-tuning may vary depending on error complexity and the model's generalization ability. While fine-tuning can improve performance, its ability to reduce radiologists' cognitive load and enhance patient care requires further real-world validation.

Despite the promising performance, we recognize several limitations to this study. First, the fine-tuning process requires

substantial computational resources and high-quality annotated radiology reports, which may pose feasibility concerns for some institutions. Indeed, the reliance on high-performance graphics processing units and high-quality annotated data for effective model fine-tuning and inference could limit the widespread adoption of these advanced LLMs in smaller clinical settings. Second, while this study focused on detecting errors in radiology reports, its generalizability to other medical documents or clinical notes remains uncertain. Although the performance of LLMs in real-world radiology report scenarios has been validated by radiologists, these models may still face

unexpected challenges in diverse clinical environments. Third, another limitation is the potential for overfitting during the fine-tuning process. The models are specifically fine-tuned on a dataset tailored to radiology report proofreading, which may limit their adaptability to different types of errors or variations in report styles that were not covered in the training data. Finally, the errors in our constructed dataset were synthetic by design. While synthetic data have been praised for their use in addressing challenges related to data scarcity and privacy, they also raise concerns about bias (34). In our study, we ensured that the synthetic data were generated from diverse and representative samples of real-world data to minimize such bias. We also validated synthetic errors by having radiologists assess their accuracy and reliability. However, synthetic errors may not fully capture the complexity of real-world errors in radiology reports. Future work could include a systematic evaluation of how bias introduced by synthetic errors affects model performance.

In conclusion, this study highlights the potential of fine-tuned large language models (LLMs) in enhancing error detection within radiology reports, providing an efficient and accurate tool for medical proofreading. The findings show that fine-tuning is crucial for enabling local deployment of LLMs while also demonstrating the importance of prompt design in optimizing performance for specific medical tasks.

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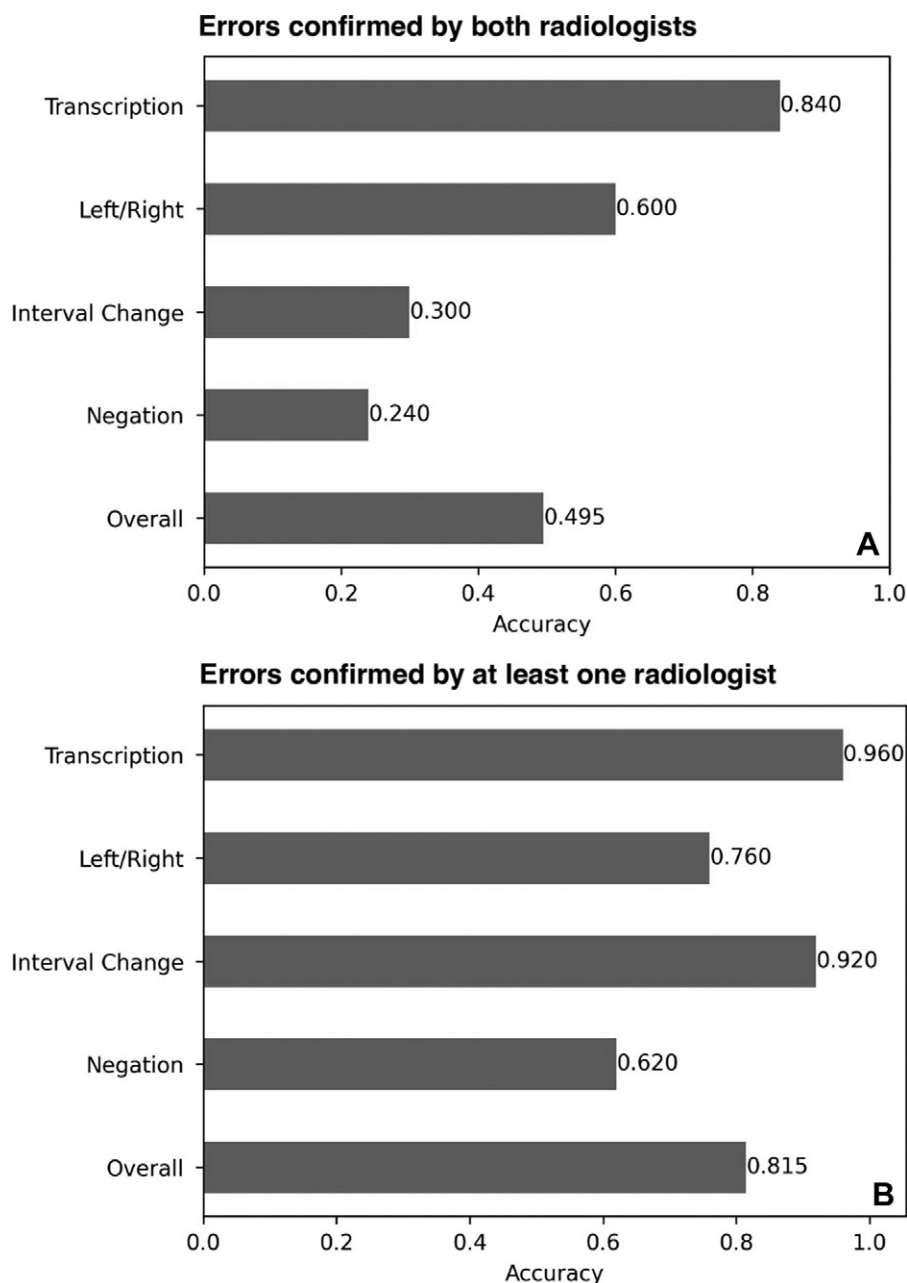


Figure 8: Bar graphs show radiologist-confirmed accuracy of error detection across different categories in radiology reports. (A) Errors confirmed by both radiologists. (B) Errors confirmed by at least one radiologist.

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